INCH POUND

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SUPERSEDING
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DETAIL SPECIFICATION

DEHYDRATORS, DESICCANT, REFRIGERANT 12

Inactive for new design after 16 February 1996

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers, nonrefillable, straight through, liquid line dehydrators suitable for with, but not limited to refrigerant type 12 (see 6.1).

1.2 Classification. Dehydrators should be of the following sizes as specified (see 6.2):

Size	Capacity in tons of refrigeration (TR).
1	2 TR (0.375-inch flare connections)
2	3 TR (0.375-inch flare connections)
3	5 TR (0.500-inch flare connections)
4	7.5 TR (0.625-inch flare connections)
5	10 TR (0.625-inch flare connections)

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in section 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MS35845 - Dehydrators, Desiccant, Refrigerant 12

(Copies of these documents are available online at https://assist.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4130

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2.3 Non-Government publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AIR-CONDITIONING, HEATING, AND REFRIGERATION INSTITUTE (AHRI)

AHRI 710 - Performance Rating of Liquid-Line Driers

(Copies of this publication are available from https://www.ahrinet.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this publication are available from https://www.asq.org.)

SAE INTERNATIONAL (SAE)

SAE-AMS2259 - Chemical Check Analysis Limits Wrought Low-Alloy and Carbon Steels

(Copies of this publication are available from https://www.sae.org.)

UNDERWRITERS LABRITORIES (UL)

UL 207 - Refrigerant-Containing Components and accessories, Nonelectrical

(Copies of this publication are available from https://www.ul.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard product. The dehydrators delivered under this specification shall be new and the manufacturer's current commercial product, except for any changes necessary to comply with specification requirements.

3.2 Code and standards compliance.

3.2.1 Safety standard. The dehydrators shall conform to UL 207.

3.2.2 Capacity rating. The dehydrators shall conform to AHRI 710, except as specified herein including capacity rating.

3.3 Materials. The materials shall conform to the requirements specified herein and on MS35845. Materials not definitely specified shall be of the quality used in the manufacturer's standard dehydrators.

3.4 Design and construction. The dehydrators shall consist of a shell containing a desiccant and an external flare connection at each end. Construction shall be as specified herein and in accordance with MS35845. The flare connection ends shall be capped and sealed against contamination during shipment and storage.

3.5 Desiccants. Desiccants shall be as specified on MS35845.

3.6 Marking.

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3.6.1 Special marking. The inlet end or the direction of flow shall be clearly stamped, etched, or embossed on the dehydrator or shown by the manufacturer's standard marking.

3.6.2 Identification. The manufacturer's name or easily identifiable trade name shall be clearly stamped, embossed, or etched on a securely attached corrosion-resistant metal plate or indicated on a pressure-sensitive label or decalcomania suitably waterproofed and attached to the shell with an adhesive.

3.7 Finish. The dehydrators shall be finished as specified on MS35845..

3.8 Mechanical properties.

3.9 Workmanship. The finished dehydrators shall be free from burrs, slivers, rough edges and malformations.

4. VERIFICATION

4.1 Conformance inspection. Inspection of the dehydrators is classified as conformance inspection (see 4.2 through 4.2.2.2).

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in ASQ Z1.4, except where otherwise indicated.

4.2.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The lot shall consist of all dehydrators of one size offered for inspection at one time. The sample unit shall be one dehydrator.

4.2.2.1 Visual examination. The dehydrator shall be examined for defects in table I. The inspection level shall be II.

TABLE I. Classification of defects.

Examine	Defect	Classification Major	Minor
Design, construction and workmanship, general	No flare connection, flare connection not capped or sealed	X	
Finish	Steel not dipped, plated, coated, or otherwise treated to resist corrosion	X	
	Sharp edges, slivers, or burrs	X	
Marking	Inlet and or direction of flow not clearly stamped, etched, or embossed on the dehydrator or not indicated on decalcomania	X	
Identification marking	Omitted, incorrect, illegible, not attached with adhesive, not specified type of plate or material, not etched, stamped or embossed, whichever required		X

4.2.2.2 Dimensional examination. The dehydrator shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2.

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5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from then managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The dehydrators covered by this specification are primarily intended for use as a component of Military field-type refrigeration equipment. Dehydrators are used for the purpose of collecting and holding within the desiccant, water in the system in excess of the amount which can be tolerated in the circulating refrigerant.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.
b. Size required (see 1.2).
c. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Flare connection
Liquid line
Tons of refrigeration

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - YD
Air Force - 11
DLA - CQ

Preparing activity:
DLA - CQ

(Project 4130-2026-009)

Review activities:
Army - CE, MD
Navy - MC
DLA - GS

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